CampaignForge Orchestrator

Autonomous Multi-Turn Agentic CRM Messaging & Workflow Architecture | Pitch & Capability Guide

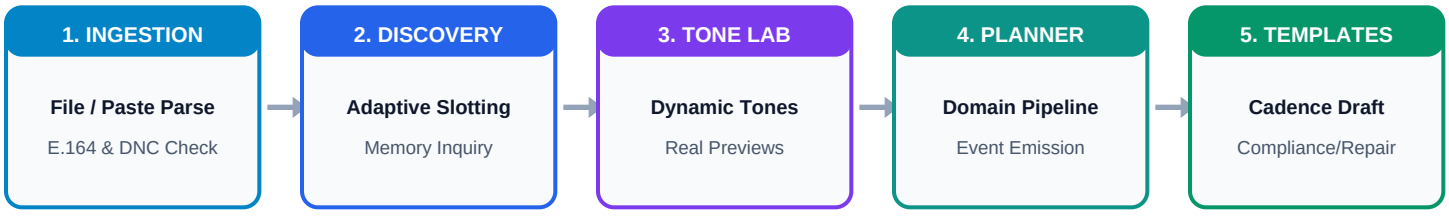
Executive Overview: CampaignForge is an adaptive, non-hardcoded AI agent that transforms raw business concepts into production-grade CRM messaging campaigns. Built on zero-hardcoded dynamic discovery, the agent autonomously determines domain requirements, cleans and maps contact lists (with phone normalization & suppression), proposes context-authentic message tones, engineers customized pipeline execution plans, and generates strictly compliant messaging sequences with verified personalization tag replacements.

1. Core Architectural Pillars & How The Agent Works

Pillar	Agent Mechanism	Business Value
Zero Hardcoding	Slots, tones, questions, and workflows are dynamically derived at runtime by the LLM from the user's specific domain (Retail, Dental, SaaS, NGOs, Education). Prompts reside in editable markdown files.	Infinitely adaptable across industries with zero code rewrites.
Pluggable Ingestion & Sanitization	Auto-detects Name, Phone, and arbitrary columns. Enforces E.164 phone normalization, deduplication, suppression filtering, and dismisses invalid files.	Clean data hygiene, zero delivery failures, and compliance with DNC registries.
Durable Workspace Memory	Persistent SQLite storage tracks brand identity, past campaigns, preferred tones, and rules across sessions.	Never repeats questions; builds contextual brand intelligence over time.
Dual-Layer Safety & Governance	Code-level LLM classification guardrails evaluate all requests and templates against a strict safety policy, failing closed on violations.	Brand protection, ethical compliance, and anti-abuse enforcement.
Strict Compliance & Self-Repair	Word limit ceiling (≤320 words), mandatory opt-out text, business naming, and tag validation with automatic repair loops.	Guaranteed deliverability under telecom regulatory limits (TCPA, TRAI, GDPR).

2. End-to-End Orchestrator Workflow Diagram

Visual representation of the agent's deterministic, multi-turn lifecycle pipeline:



3. Full Capabilities & Orchestrations Possible by the Agent

Pipeline Stage	Autonomous Actions Executed by Agent
Stage 1: Contact Intake & Hygiene	• Pluggable CSV, XLSX, TSV, and pasted lines parsing. • Automatic column header detection (Name, Phone, custom fields). • Normalization to E.164 (+977..., +1...) via Google phonenumbers library. • Deduplication by phone number; suppression database lookup against data/suppression.json. • Immediate rejection of corrupted files with 0 valid numbers.
Stage 2: Adaptive Discovery & Memory	• Domain classification (Retail, Healthcare, NGO, B2B SaaS, Education). • Dynamic slot extraction: business name, campaign goal, audience, CTA, link, messages count, cadence. • Memory integration: Proactively checks past workspace campaigns and asks whether to reference past donations/actions. • 1-2 focused questions at a time; states assumptions when user says 'you decide'.
Stage 3: Tone Exploration & Agreement	• Proposes 3-5 domain-tailored candidate tones (e.g. 'Gentle & Reassuring' for clinics; 'High-Energy Hype' for retail). • Crafts live preview examples with real campaign details. • Supports user refinement loops (e.g. 'make it punchier' or 'less formal') until locked in.
Stage 4: Dynamic Planning & Event Streaming	• Synthesizes customized orchestrator steps tailored to the exact campaign. • Emits standard <pipeline_event> SSE events driving live animated timeline UI. • Status progression: active → processing → done → waiting_for_input.
Stage 5: Compliant Template Generation	• Generates multi-touch message sequence (Announcement → Follow-up → Final Call). • Strict ceiling validation (≤320 words). Auto-regenerates with repair prompt if breached. • Guaranteed mandatory opt-out ending ('Reply STOP to unsubscribe'). • Real recipient previews replacing {{customer_name}}, {{phone}}, and custom row columns. • Individual message editing (═■ Edit) and LLM-assisted regeneration (■ Regenerate).
Stage 6: Enterprise Audit & Export	• SQLite relational database mirror + append-only JSONL files. • Full prompt, raw response, token usage, latency (ms), and safety verification logs. • One-click export of approved Campaign Package JSON & Full Session Run Log.

4. Real-World End-to-End Example Conversation

Below is an actual conversation script demonstrating memory recall, contact integration, and template generation:

Turn	Speaker	Dialogue & Agent Output
1	**User**	*"Hello! We want to run a follow-up campaign for flood relief in Nepal."*
2	**Agent**	**Welcome back!** I see that **Hope for Nepal Foundation** previously ran the *Nepal Flood Relief Fund* with us. Before we draft the sequence: **Would you like this new message to specifically acknowledge past donor contributions** (e.g. *'You donated last time for Nepal flood relief and your support saved lives—will you stand with us once more?'*)? Also, please upload your donor list so I can personalize names.

3	**User**	**Uploads mock_contacts_nepal_100.csv** *"Yes, definitely mention their past support. Goal is $50 emergency medical kits. Suggested gap is 3 days."*
4	**Agent**	**[OK] 105 clean contacts sanitized & ready** (0 duplicates, 0 suppressed). *** Campaign Configured:** Hope for Nepal Foundation | Goal: $50 Medical Kits | Past donor reference: Enabled. **Candidate Tones Proposed:** • **[Tone 1] Urgent & Dignified**: *'Dear {{customer_name}}, you helped us provide emergency aid during the last floods. As monsoons return, 120 families need medical kits...'* • **[Tone 2] Community Solidarity**: *'Hi {{customer_name}}, collective action saved lives in Nepal last year...'* Please select a tone to plan the pipeline.
5	**User**	**Clicks "Urgent & Dignified"**
6	**Agent**	*** Pipeline Planned & Executed:** [OK] Sequence structured (Day 1 -> Day 4 -> Day 7) [OK] 3 compliant templates drafted (<= 320 words, opt-out verified) **Live Recipient Preview (Message 1):** *** Ramesh Adhikari** (+97.....0000 | City: Kathmandu | Tier: Supporter): *"Dear Ramesh Adhikari, we hope this finds you well. As a valued supporter of Hope for Nepal Foundation, your past generosity brought relief to families in need. Monsoons have once again displaced communities, and we urgently need $50 medical supply kits... Reply STOP to unsubscribe."* [Edit] [Regenerate] [Approve Campaign]

Summary: CampaignForge delivers end-to-end automation without sacrificing human oversight or safety. It replaces static forms with intelligent, contextual conversation and produces audit-ready, compliant campaigns in seconds.